

A Comparison Study on Stress Relief in VR

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ABSTRACT

In modern life, people encounter various stressors, which may cause negative mental and bodily reactions to make them feel frustrated, angry, or irritated. Earlier studies have revealed that Virtual Reality (VR) experiences are effective as traditional methods in reality in stress management and reduction. However, it is still unclear how different activities in VR and their combinations could moderate the effects on stress reduction. In this work, we conduct a comparative study to investigate the effects of two virtual activities: 1) meditation in VR, 2) a smash room experience in VR, and 3) a traditional meditation method of *Sitting in Silence* in the real world. The results show that VR meditation has the highest short-term effects on temporal stress, followed by VR smash room and *Sitting in Silence*.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI

1 INTRODUCTION

People encounter various stressors, such as anxiety-inducing tasks, work deadlines, or discomfort in social situations. They may cause negative mental and bodily reactions to make people feel frustrated, angry, or irritated. Different approaches to manage and reduce stress or anxiety are found to be useful and effective for stress relief, such as meditation, taking a rest, walking around nature, or even breaking objects [2].

For decades, Virtual Reality (VR) has been widely used in the context of mental (and physical) therapy given the potential benefits as a psychotherapy treatment tool [3]. For instance, Tarrant et al. [5] and Kaplan-Rakowski et al. [1] investigated the effect of VR-based meditation by comparing it to conventional audio- and video-based meditation methods. They reported that VR meditation provided better or comparable experiences to traditional meditation methods. However, there is still a dearth of research investigating the effects of different stress relief methods in VR and their combinations on stress reduction. In this work, we conduct a formal user study ($N = 40$) to evaluate two stress-management methods in VR environments: 1) meditation and 2) a smash room experience, while comparing them with 3) a traditional *Sitting in Silence* method in the real world setting. Our results show that all three methods reduce the level of stress, and the meditation experience in VR is the most effective method compared to the other two methods.

2 EXPERIMENT

2.1 Stress Relief Methods

Our study compares *Sitting in Silence*, *VR Meditation*, and *VR Smash Room* as stress-management methods.

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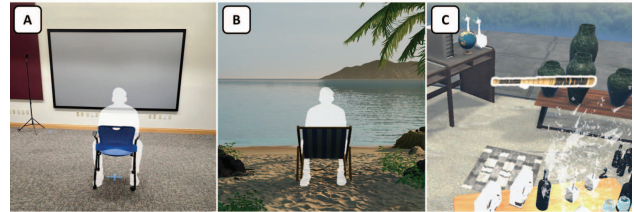


Figure 1: (A) *Sitting in Silence* is a mindfulness observation method asking users to sit in a chair and relax with no guided voice. (B) *VR Meditation* presents an immersive beach. Users can take a rest sitting in the chair and explore the scene. (C) *VR Smash Room* is designed for users to do the physical activity by smashing virtual objects.

Sitting in Silence: This is one of the traditional meditation method in the real world. As the name indicates, *Sitting in Silence* is taking a rest in a silent space (Fig. 1A).

VR Meditation: Participants in the VR meditation condition experience a virtual beach scene (Fig. 1B). We utilize *Meditation VR*¹, a free application developed to provide VR meditative experiences. It satisfies four of the five criteria that Zaharuddin et al. [7] suggested for better virtual meditation spaces: visual attractiveness, realism, user comfort, and navigation methods.

VR Smash Room: It is a virtual room where people can destroy and break objects (Fig. 1C). The room includes VR objects such as windows, televisions, vases, desktop computers, wine bottles, etc. The participants can break these items by swinging a VR controller, which is represented as a baseball bat in the VR scene.

2.2 Participants

A total of 40 participants (13 male, 27 female; age mean: 19, range: 18–22) were recruited through the university's student participants recruitment system, SONA. All participants had 20/20 (or corrected 20/20) vision and had no impairments for the use of VR devices. The study took approximately an hour, and they earned 1 SONA credit as stipulated based on the university SONA policy.

2.3 Apparatus & Study Space

The study was performed using an HTC Vive Pro Eye headset with a Vive wireless adapter and one Vive controller. The headset has a viewing angle of 1440×1600 for each eye and a refresh rate of 90 Hz. Two virtual scenes (i.e., the VR smash room and meditation scenes) ran on a Windows 10 desktop with an Intel Xeon W-2245 CPU (3.90 GHz), 64 GB of RAM and an Nvidia GeForce RTX 3090 graphics card. The tracking space for the VR scenes was $4m \times 4m$.

2.4 Procedure

Upon the arrival on the study place, a participant was asked to read and sign the informed consent form to participate in the study (IRB #12971). Next, the participant was asked to complete a stress assessment questionnaire, State-Trait Anxiety Inventory for Adults (STAI) [4]. The details about STAI are described in the next section. This first STAI assessed the participant's initial stress level.

Following that, the participant was asked to perform a Paced Auditory Serial Addition Test (PASAT) three times. The PASAT is

¹https://store.steampowered.com/app/1301850/Meditation_VR/

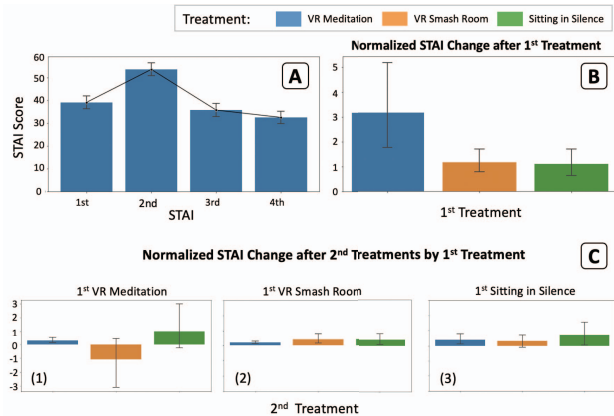


Figure 2: The results along with 95% CI error bars. (A) The average STAI scores by the steps. (B) The average values of the normalized STAI changes after the first treatments. (C) The average of the normalized STAI changes after the second treatment by the first treatment. (1), (2), and (3) shows the second treatment effects after VR meditation, VR smash room, and Sitting in Silence respectively.

a cognitively challenging task to induce the increase of stress [6]. After PASAT, the participant was asked to complete the second STAI.

Next, the participant experienced one of three interventions (i.e., Sitting in Silence, VR meditation, and VR smash room) for 5 minutes, and completed the third STAI. Following the first intervention, the participant was engaged in one of three interventions for another 5 minutes and completed the fourth STAI. The first and second interventions allowed to be redundant (e.g., two smash rooms), and their sequences were counter-balanced.

2.5 Stress Measure: Level of Anxiety

Stress is typically caused by an external and existent source, and anxiety could be considered as a different type of stress that remains even if the stress factor is gone. To evaluate the stress level after each procedure step, we employ STAI. It is a self-report measurement method on a 4-point Likert scale. It has two forms of Y-1 and Y-2. The Y-1 form evaluates users currently feeling, and the Y-2 form is a set of statements to evaluate the users' general feelings. In this study, we use only the Y-1 form. It has a total of twenty statements like "I feel calm" and "I am tense". The level of anxiety is decided by calculating the total score.

3 RESULTS

We first analyzed the STAI scores to investigate the within-subjects effects of our interventions, such as PASAT and two treatments by using one-way analysis of variance (ANOVA) at the 5% significance level. To compare the relative stress reduction effects of three treatments, we decided to analyze the normalized amount of STAI changes because the sensitivity to stress varies individually. The normalized changes were calculated by dividing the actual STAI score changes during the first or second treatment by the amount of initially increased stress in the PASAT task. They were also analyzed by using the one-way between-subjects ANOVA at the 5% significance level. Then, we performed pairwise comparisons as the post-hoc test at the same 5% significance level.

Here, we first report the comparison result of the average STAI scores (Fig. 2A). It clearly shows that the interventions had an impact on STAI results ($p < .001$, $F = 76.6$, $\eta_p^2 = .663$), with scores rising after PASAT and falling after two treatments. The pairwise comparisons disclose that PASAT ($m = 55.1$) significantly increased the participants' STAI score by 16.9 compared to the previous stage

($m = 38.2$, $p < .001$). The first and second treatments reduced the stress level by 19.5 and 3.55 on average but no statistical significance was found.

Next, the results of the between-subjects ANOVA regarding the normalized STAI changes and treatments are reported. Fig 2B shows the changes along with their 95% CI after the participants had first experienced one of three treatments. A significant difference between them are found ($p = .028$, $F = 3.95$, $\eta_p^2 = .176$). In its pairwise comparisons, the VR meditation group ($m = 3.16$) has a significantly higher amount of stress reduction than VR smash room ($m = 1.17$, $p = .021$) and Sitting in Silence ($m = 1.09$, $p = .018$). However, no significant difference is disclosed between the VR smash room and Sitting in Silence ($p = .926$). We then examine the order effect of the treatments. It is evaluated separately for each first treatment (Fig. 2C). We found no significant differences between the second treatments after VR meditation ($p = .277$). The result shows that practicing VR smash room after VR meditation increased the participants' stress on average but no statistical significance is found. No significant differences between the treatments as a second treatment after VR smash room ($p = .689$) and Sitting in Silence ($p = .675$).

4 CONCLUSION

In this work, we conducted a formal user study to investigate the single and combinatorial effect of the three treatments on stress relief. We observed the stress levels were significantly changed by following our study procedures. The results confirmed that PASAT increased the stress level of the participants and all the treatments could reduce their stress levels. Among the three treatments, VR meditation had the biggest stress relief effect as the first treatment compared to VR smash room and Sitting in Silence. Most of the second treatments had beneficial effects on reducing stress, but we discovered an interesting combination of VR meditation and smash room, which increased the participants' stress levels. However, we found a few limitations in our study, e.g., using STAI does not allow us to monitor the stress level in real-time while a participant is experiencing a treatment. Additionally, only a small group between the ages of 18 and 22 was studied in this work. We expect that using specialized metrics, such as biomarkers, and broadening the demographic group will enable us to better understand how VR affects stress reduction.

REFERENCES

- [1] R. Kaplan-Rakowski, K. R. Johnson, and T. Wojdyski. The impact of virtual reality meditation on college students' exam performance. *Smart Learning Environments*, 8(1):1–15, 2021.
- [2] P. M. Lehrer, R. Carr, D. Sargunary, and R. L. Woolfolk. Stress management techniques: Are they all equivalent, or do they have specific effects? *Biofeedback and Self-Regulation*, 19(4):353–401, dec 1994. doi: 10.1007/BF01776735
- [3] M. J. Park, D. J. Kim, U. Lee, E. J. Na, and H. J. Jeon. A literature overview of virtual reality (vr) in treatment of psychiatric disorders: recent advances and limitations. *Frontiers in psychiatry*, 10:505, 2019.
- [4] C. D. Spielberger. State-trait anxiety inventory for adults. 1983.
- [5] J. Tarrant, R. Jackson, and J. Viczko. A feasibility test of a brief mobile virtual reality meditation for frontline healthcare workers in a hospital setting. *front. Virtual Real.* 3: 764745. doi: 10.3389/fvrvir, 2022.
- [6] T. N. Tombaugh. A comprehensive review of the paced auditory serial addition test (pasat). *Archives of clinical neuropsychology*, 21(1):53–76, 2006.
- [7] F. A. Zaharuddin, N. Ibrahim, M. E. Rusli, E. M. Mohd Mahidin, and A. M. Yusof. Factors to consider when designing a virtual environment to treat stress. In *International Visual Informatics Conference*, pp. 36–47. Springer, 2019.